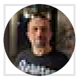
Time Series with Konrad: episode 5

Survival analysis for time series



KONRAD BANACHEWICZ

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In most time series problems, we ask *what happens next*. Survival analysis asks a slightly different question: **when will it happen?** That difference is more important than it sounds.

What do churn modeling, equipment failure, customer onboarding, and ticket resolution have in common? Timing *is* the signal. Knowing that something will eventually happen is often less useful than knowing how long it usually takes, who is at risk right now, and how that risk changes over time.

Survival analysis grew out of medical statistics (*hence the terminology*), but the ideas translate naturally to modern time series problems: incomplete histories, rolling observation windows, and decisions that must be made *before* all outcomes are known. In this episode, we'll build survival analysis from the ground up, but with a time-series focus. We start with simple time-invariant tools, then move toward models that handle covariates and non-linear effects, ending with tree-based survival models.

📓 *All code lives in the companion notebook:* https://github.com/tng-konrad/time-series-with-Konrad/blob/main/m05-survival-analysis.ipynb

Crash intro to survival analysis

So far, we've mostly worked with time series where the past is fully observed: **we know what happened and when**, and we build predictions from that complete history. Unfortunately, that's not how many real-world problems work. Two key questions often come up:

- Do we need to act before all data is observed?
- Does the timing of an event matter?

If the answer is *yes* to both, survival analysis is a good fit. This involves a bit of a shift in mindset: we're no longer predicting the next value, but reasoning about *when a sequence will end*.

Survival analysis is a group of statistical methods designed to model **time to event:** how long it takes for something of interest to happen. In practice, we must close our observation window at some point to make decisions, which means we can't wait for all events to occur. This means that some observations are **censored**.

For censored subjects, the event hasn't happened yet, so we only know that the time-to-event exceeds a certain duration. This makes the outcome two-dimensional: **whether** the event occurred and **when** it occurred.

Survival models support both prediction and causal inference. They describe probability distributions of events happening at / by / after a given time, and they quantify how different factors influence event timing. Survival analysis is used to:

- **Quantify and visualize duration distributions** - define expectations, baselines, and thresholds
- **Estimate survival curves for individual subjects** - prioritization and targeted intervention

What are some examples of problems we can treat with survival analysis apparatus?

- People / customers: churn, first purchase, employee tenure
- Systems: equipment failure, bugs, inventory depletion
- Finance: loan repayment, break-even revenue

So much for a (quick) intro - how do we make the idea more precise? I'm glad you asked.

Basic toolkit

We need three main building blocks to start applying survival analysis:

- the random variable *T* representing the time of event of interest - **random** because we do not know when things will happen
- survival function *S* - probability of an event taking place later than time t
- hazard function *h* - instantaneous risk

Think of each customer (*or machine, or ticket - see list in the prev paragraph*) as a *time series that ends with an event* **-** except many of them haven't ended *yet*. These three objects describe the same phenomenon: when the event happens, how many survive, and how risky things are right now.

Censoring

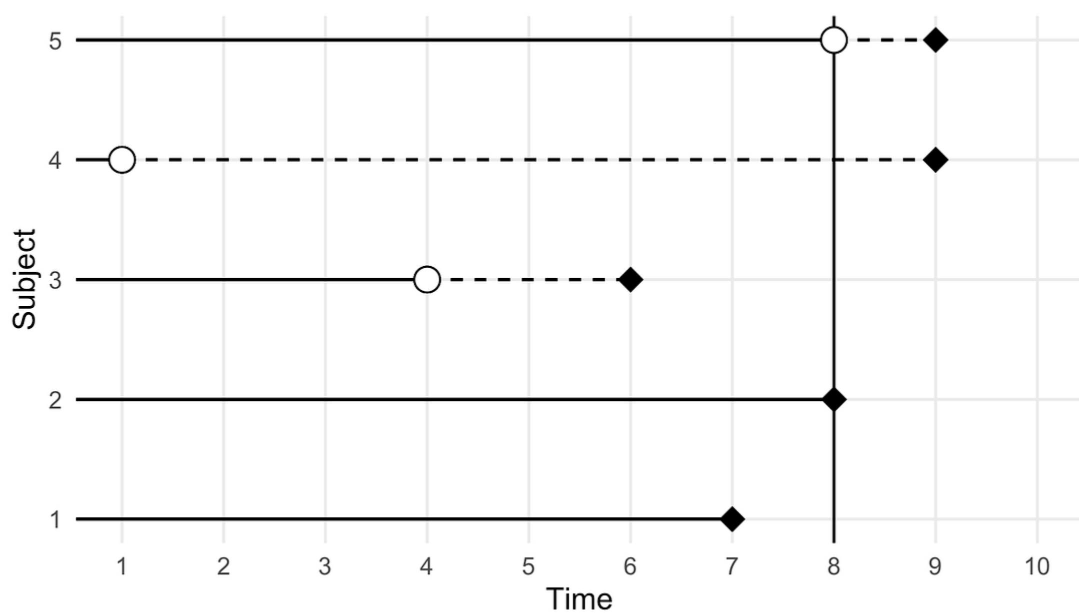
So far so good - but what makes survival analysis special is **censoring**: the NA of survival analysis. Observations are censored when the information about their survival time is incomplete:

- Up to this point in time, the event has not happened.
- We don't know what could've happened after the observation period, but *the absence of event is informative about the risk*

There are 3 types of censoring: right, left and interval censoring:

- **Right censoring**: The most common type of censoring: occurs when the survival time is incomplete at the end of the observation period
- **Left censoring** occurs when we cannot observe the time when the event occurred
- **Interval censoring**: We performed testing at *t1* and *t2* - negative outcome first, positive second - we only now the event happened at some point between t1 and t2

That information is still valuable, but it breaks standard regression approaches that assume we observe the full outcome for everyone. As usual with these things, picture is worth a thousand words:



- Each horizontal line represents one subject
- When we see a black diamond, the event of interest has occurred. When the line ends without a diamond, observation stops before the event happens.

This diagram shows why censoring is not “missing data” in the usual sense (*useful as the analogy is*): **every subject contributes information up to the point we stop observing them**. Even when the event hasn’t happened yet, we still know how long it *did not* happen - and that constrains the risk going forward.

Survival analysis is built to use exactly this partial information, while ordinary regression methods *(OLS and all the other linear model we know and love ;-)* are not.

Survival function

Survival function is given by:

$$S(t) = P(T > t)$$

Interpretation: probability of the event of interest **not occurring** by time t. As an example, for churn on monthly data level S(12)=0.8 means 80pct of customers are still active (= have not churned) after one year.

Survival estimation comes in three flavors:

1. non-parametric: survival probabilities are dependent on time. **Kaplan-Meier**
2. semi-parametric: Cox regression
3. parametric: survival times follow specific probability distribution (e.g. Weibull, exponential, lognormal)

We’ll start with the simplest case: the non-parametric Kaplan–Meier estimator.

KM estimator

Kaplan-Meier estimator:

- non-parametric estimate of the survival function
- break the estimation into steps depending on observed event times

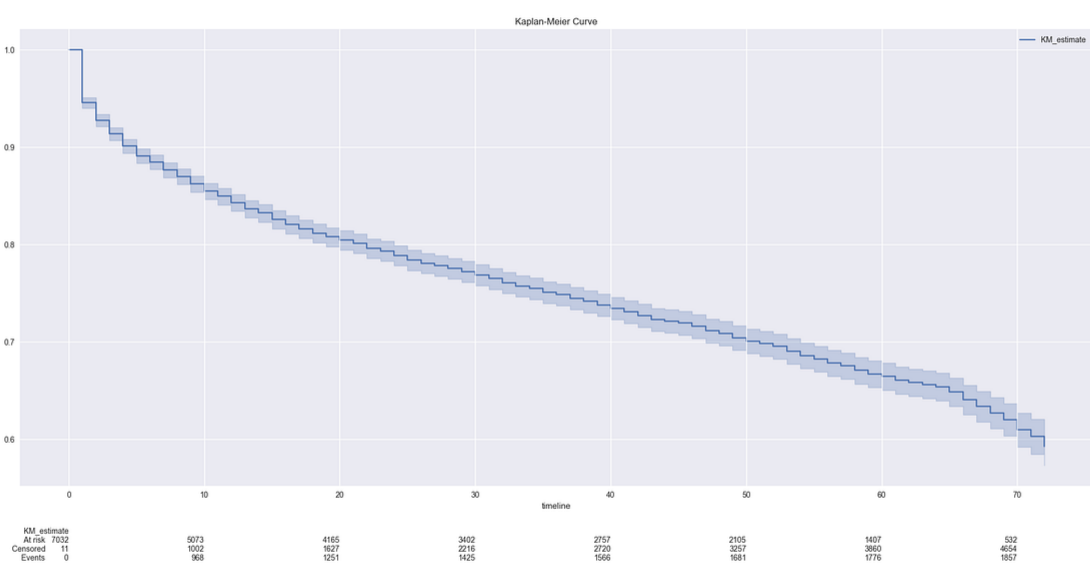
$$\hat{S}(t) = \prod_{i:t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

where *ni* is a number of individuals who are at risk at time point *ti* and *di* is a number of subjects that experienced the event at time *ti*.

The Kaplan–Meier estimator is the simplest way to estimate a survival curve in the presence of censoring: it works by updating the survival probability **only at times when events occur**, which is why the resulting curve is step-shaped rather than smooth.

Assuming you are still here after the - necessary! - theoretical introduction, let's see what we can do about applying survival analysis to a churn dataset (Kaggle Dataset: https://www.kaggle.com/datasets/blastchar/telco-customer-churn).

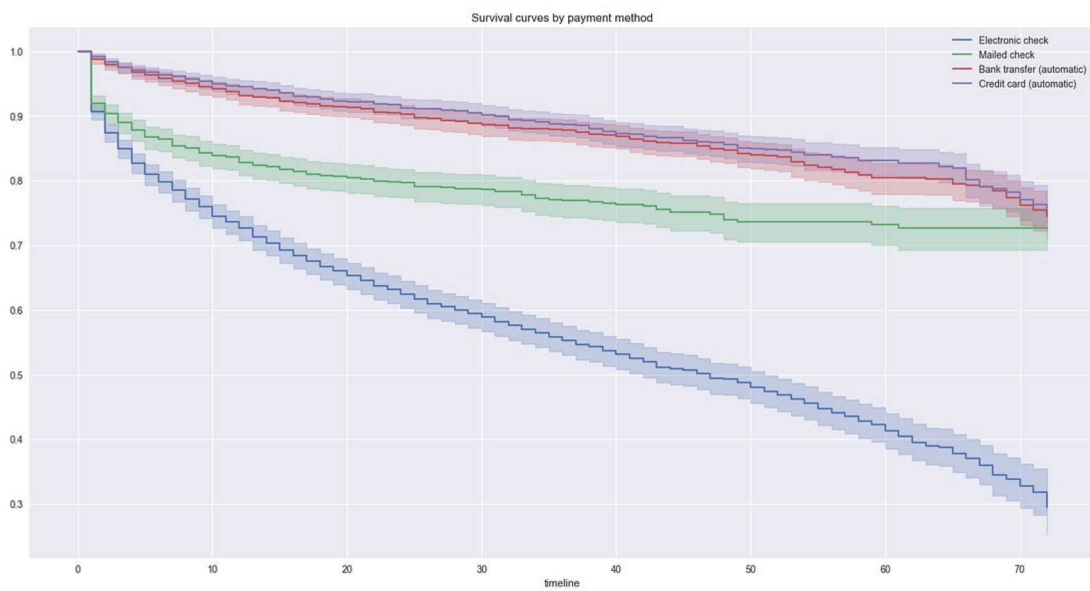
This is the KM estimator of the survival curve:



This graph shows the estimated probability that a customer is still active as a function of tenure:

- The x-axis shows customer tenure in months
- The y-axis shows the probability a customer has *not* churned yet
- Each drop corresponds to observed churn events
- Shaded bands represent uncertainty, which grows as fewer customers remain under observation

What if we'd like to get more insight into behavior per subset of the population (= level of a categorical feature)?



Here we see clear separation between payment methods. Some groups churn much faster early on, while others decline more slowly over time. Importantly, these differences emerge **without fitting a predictive model** - they are visible directly in the data.

The log-rank test simply formalizes this visual comparison. It answers a narrow question: *are these curves different beyond what we'd expect by chance?*

		test_statistic	p	-log2(p)
Bank transfer (automatic)	Credit card (automatic)	0.87	0.35	1.51
	Electronic check	510.04	<0.005	372.74
	Mailed check	51.07	<0.005	40.03
Credit card (automatic)	Electronic check	539.74	<0.005	394.21
	Mailed check	64.82	<0.005	50.11
Electronic check	Mailed check	152.46	<0.005	113.93

We ran the formal test mostly for completeness and demonstration purposes - in everyday EDA type of work, visuals are fit for purpose.

K-M summary:

- (+) flexible, minimal assumptions
- (+) fast
- (-) no covariates incorporation → prediction :-(
- (-) not smooth

Can we do better?

Survival curves tell us *how many are left*. Hazard rates tell us **who is at risk right now and answer a more** local question:

Given that the event hasn't happened yet, how likely is it to happen right now?

You can think of it as an "instantaneous churn pressure" - mathematically, it is defined as

$$\lambda(t) = -\frac{S'(t)}{S(t)}$$

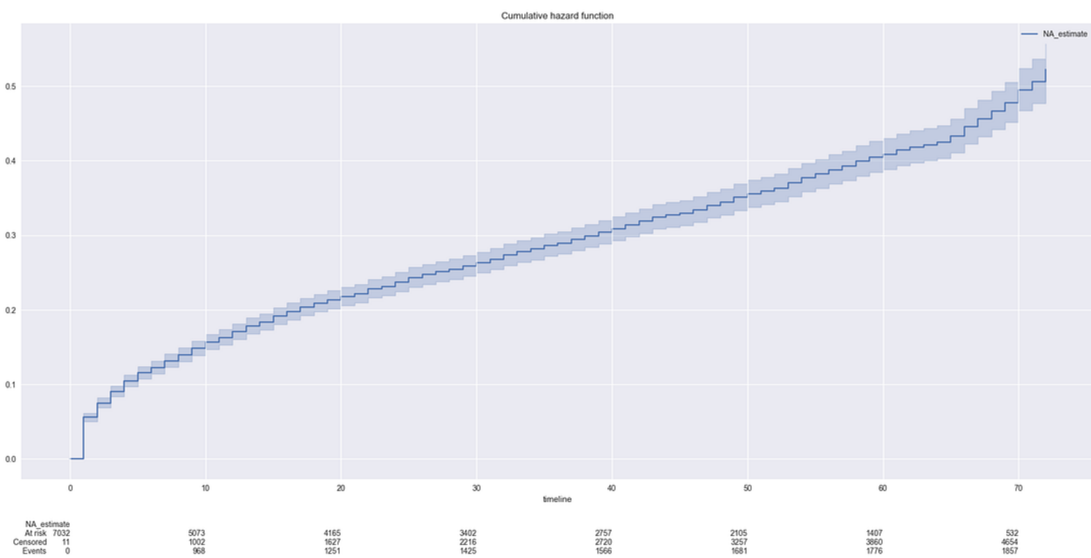
The cumulative hazard is the total risk accumulated over time:

$$\Lambda(t) = \int_0^t \lambda(s)ds = -\log S(t)$$

The Nelson–Aalen estimator gives a non-parametric estimate of cumulative risk and serves as a building block for Cox model - regression technique for censored data :

$$\hat{\lambda}(t) = \sum_{i:t_i \le t} \frac{d_i}{n_i}$$

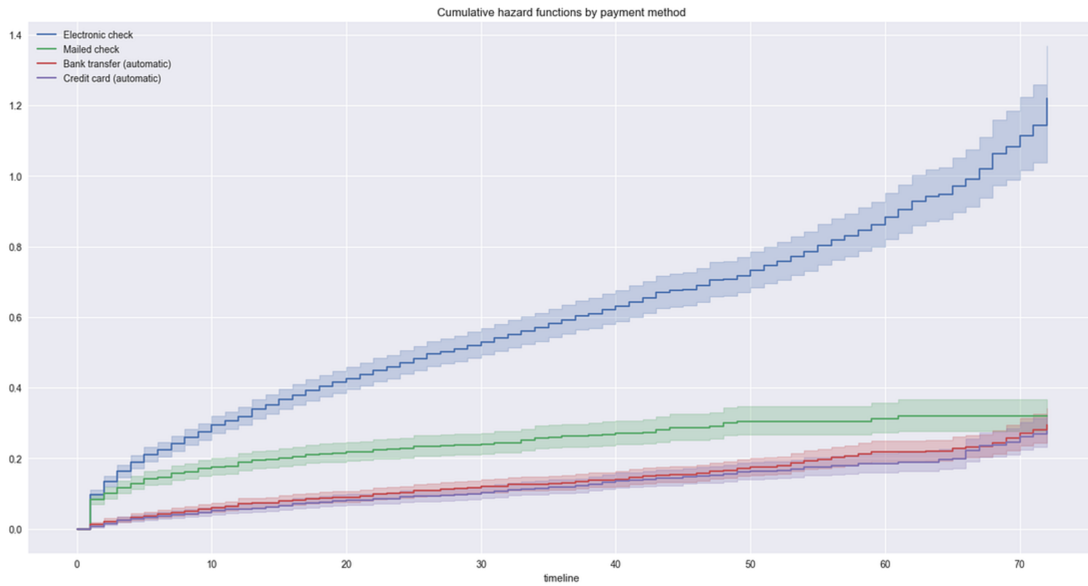
- less popular (due to interpretation)
- measures total amount of risk accumulated up to time *t*
- intuition: expected events if non-absorbing



This plot shows the cumulative hazard estimated using the Nelson–Aalen estimator. The survival curve answers “how many are left” - the cumulative hazard tracks **how much total risk has accumulated over time**. It is always increasing: risk can pile up, but it cannot be undone.

Cumulative hazard is not exactly intuitive by itself, but it becomes extremely useful as a building block for regression models - especially

Cox regression, which operates on hazard, and not on survival probabilities.



When we plot cumulative hazard by subgroup, differences in risk become easier to compare on an additive scale: groups with steeper curves accumulate risk faster, meaning churn pressure builds more quickly over time. These plots emphasize *how risk grows*, not how many customers remain.

This perspective is especially helpful when transitioning to models that directly parameterize hazard, such as Cox regression.

Cox regression

Cox regression connects survival analysis to regression models you already know.

Instead of modeling time directly, it models **relative risk**:

$$\lambda(t) = \lambda_0(t) \exp\left(X^T \beta\right)$$

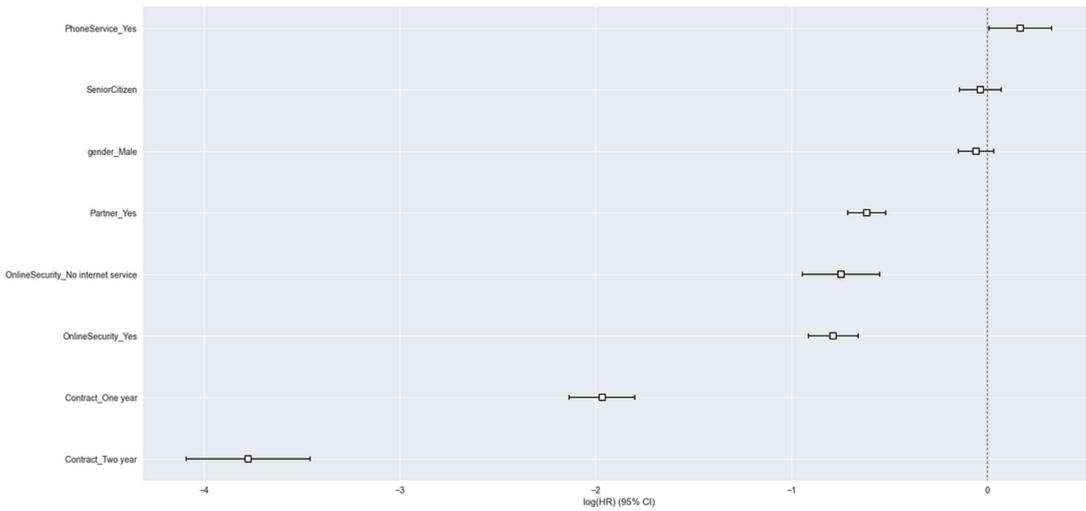
- The key idea: covariates don't change *when* churn happens, but they scale the risk up or down at every point in time.
- Covariates have a multiplying effect on the hazard rate
- Effect is constant over time
- Works with right-censored data

Performance of a Cox model is measured by Concordance index. For a survival model C-index ≈ weighted average of the area under time-specific ROC. The C-index measures how well the model ranks customers by risk. It ignores exact timing and only asks: *did higher-risk customers churn earlier?*

Intuition:

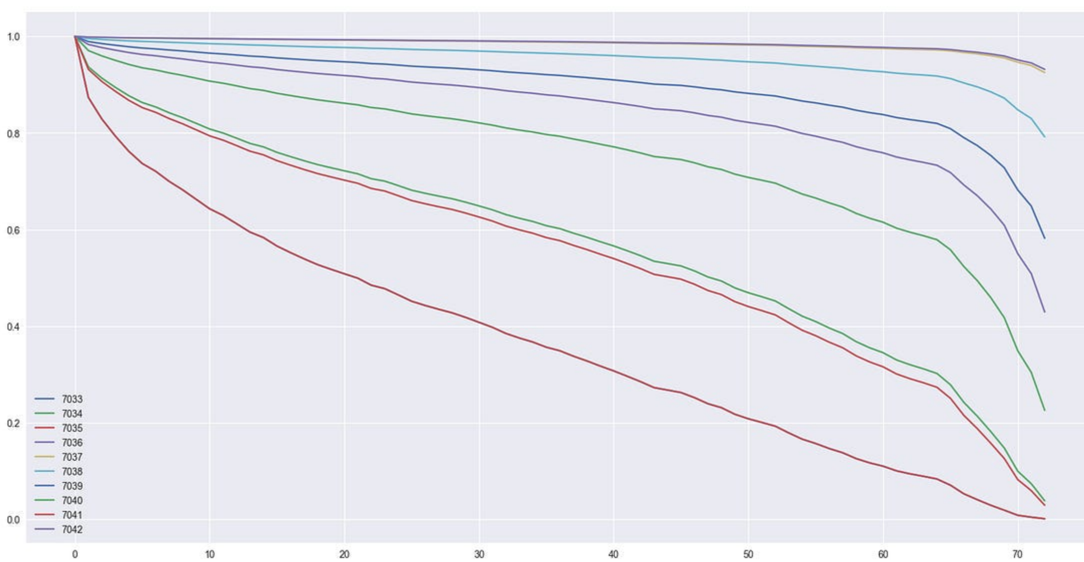
- C-index measures discrimination power
- **Only evaluates the rank**: does model predict same order of churn as really happened?
- Not impacted by value (duration)
- Necessary evil for right-censored data :-/

Cox regression estimates how each covariate scales risk relative to a baseline. Coefficients to the right increase churn risk; coefficients to the left reduce it. The horizontal bars show uncertainty - if a bar crosses zero, the effect is weak or ambiguous:



Importantly, Cox regression predicts *who is at higher risk at every point in time - not* when churn happens. **Interpretation is therefore relative, not absolute.**

Cox model produces predictions of individual survival curves:

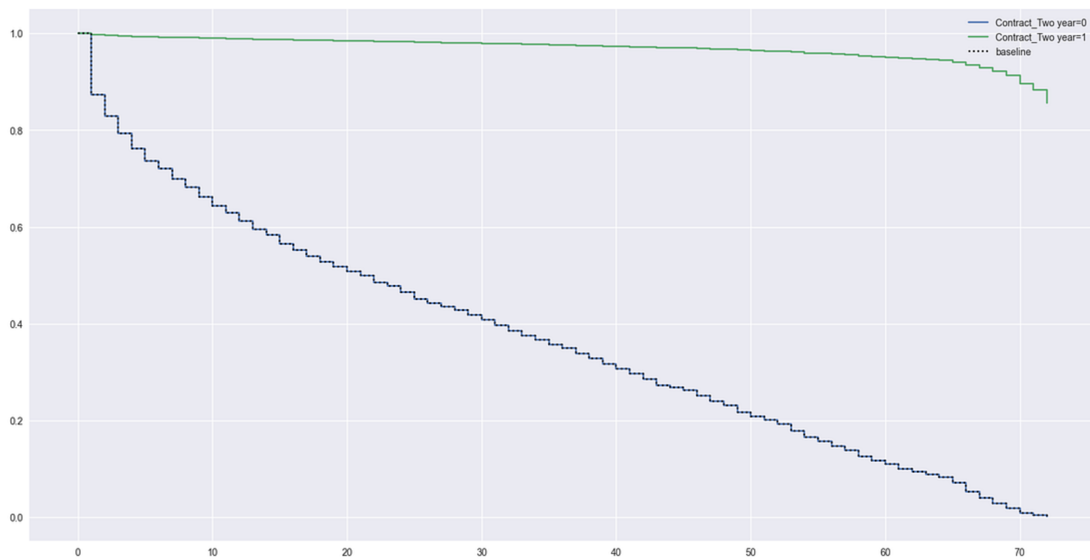


Cox regression models estimate relative risk, but you can still produce individual survival curves - by combining estimated risk with the baseline hazard. Each curve represents a different customer profile.

Customers with higher estimated risk see their survival probability decay faster over time, while lower-risk profiles decline more slowly.

These curves are best used for **ranking and prioritization**, not precise time predictions. We can also zoom in on the diagnostics:

```
cph.plot_partial_effects_on_outcome(covariates='Contract_Two year', values = [0,1])
```



Partial effects plots isolate the impact of a single covariate while holding all others constant. Here, we compare survival under different contract lengths. The key observation is that the curves remain proportional over time - the core assumption in Cox regression makes. If these curves crossed or diverged irregularly, it would signal a violation of the proportional hazards assumption.

Once we're comfortable with linear effects, the natural step is to drop linearity altogether.

Survival forest

Up to this point, we've been steadily relaxing assumptions. We started with Kaplan–Meier, which makes almost no assumptions - but cannot use covariates. Cox regression added covariates, but at a price: linear effects and proportional hazards. The natural next step is to ask what happens when those assumptions no longer hold. This is where **survival forests** come in.

A survival forest combines random forests with survival analysis. Conceptually, it replaces the single global hazard model of Cox regression with an ensemble of decision trees, each learning **local risk patterns** in different regions of the feature space - while still handling censoring.

The key improvement over Cox regression is flexibility:

- Effects do not have to be linear
- Interactions emerge automatically
- Risk does not have to scale proportionally over time

As a result, survival forests can produce survival curves that differ both in *level* (higher vs lower risk) and in *shape*. This allows the model to capture behaviors such as early churn vs late churn, saturation effects, or regime changes - patterns that Cox regression cannot represent by design.

The trade-off is interpretability. Survival forests are harder to explain coefficient-by-coefficient, but they often win when **predictive accuracy** matters more than clean parameter estimates.

Survival forests still operate in a survival setting: they predict a **full survival curve for each subject**, not a single time-to-event. This means we can answer questions like:

- "What is the probability this customer survives another 6 months?"
- "Which customers are most at risk *right now*?"

- “How does risk evolve differently across segments?”

In practice, this makes survival forests particularly useful for **prioritization, intervention timing, and scenario analysis**, rather than point predictions of when an event will occur.

Previously, we evaluated Cox regression using the concordance index. The C-index measures how well a model ranks customers by risk, but it ignores calibration: a model can rank customers correctly while still being badly wrong about *how likely* events actually are. This is where the **Brier score** becomes useful.

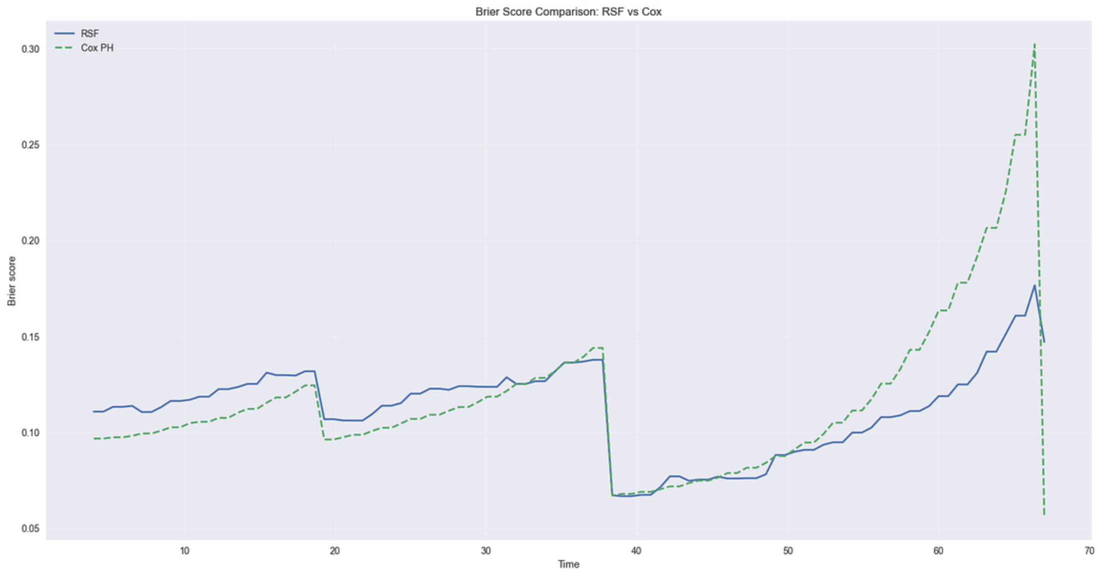
In survival analysis, the Brier score measures how close the predicted survival probabilities are to reality at each point in time. Concretely, it computes the average squared difference between:

- The observed survival status (event or no event), and
- The predicted survival probability at time *t*

Lower values are better. The Brier score penalizes:

- overconfident predictions (*sort of like log-loss blows up if you are very confident AND wrong*)
- poorly calibrated survival curves
- models that get the timing roughly right, but the probabilities wrong

This makes it especially relevant when survival probabilities are used directly in decisions: for example, when triggering interventions based on estimated risk thresholds



This plot compares the Brier score over time for Cox regression and the survival forest. Across most of the time horizon, the survival forest achieves a lower Brier score than Cox regression. This indicates that its predicted survival probabilities are closer to observed outcomes - *better ranked **and** better calibrated.*

The gap tends to widen in the second half of the data, where uncertainty is higher and modeling assumptions matter more. This is where Cox regression’s linear and proportional assumptions become restrictive, while survival forests can adapt to changing risk dynamics.

Cox regression is a strong baseline: fast, interpretable, and well suited for understanding *relative risk*. It works best when effects are roughly linear, proportional over time, and when ranking subjects by risk is the primary goal.

Survival forests trade interpretability for flexibility. They shine when risk is non-linear, interactions matter, and when **accurate survival probabilities** are needed for decisions (and not just rankings). The improvement in Brier score reflects exactly that: better-calibrated predictions across time.

In practice, this isn't an either–or choice (also: No Free Lunch theorem) . Cox models are often the right starting point; survival forests are a natural next step when assumptions begin to strain and prediction quality becomes the bottleneck.

Closing time

Survival analysis fits naturally into the time series toolkit. Whenever the *timing* of an event matters, and whenever you're forced to act before observing the full future, survival methods give you a principled way to reason under uncertainty.

In this episode we stayed classical: Kaplan–Meier, Cox models, and trees. Extensions to time-varying covariates and deep learning exist - but the ideas you've seen here already cover a surprisingly large fraction of real-world use cases.